❶ Read: KBo 44.52, an orphan (CTH 470)

1′ x-*a-ḫi*-
2′ NINDA.SIG x x
3′ EN SÍSKUR-*ma* *AN*
4′ LÚSIMUG-*ia* *ku-e* x
5′ LÚḪAL *ša-ra-a* *da-a-i*
6′ x$^{\text{HLA}}$-*ma-aš-ša-an* *ku-i-e-eš* *A-NA* GIŠBANŠUR
7′ x *da-aš-ke-ez-zi* *na-at-kán* *ḫa-aš-ši-i* *zi-Ø-Ø-Ø-Ø*
8′ *-in-ni* *zi-ik-ke-ez-zi* EN SÍSKUR-*ma*
9′ *na-an-kán* EN SÍSKUR *A-NA* D*IŠTAR* *ši-pa-Ø-Ø*
10′ *šu-up-pa* *ḫu-u-i-šu* *da-an-zi* UZUGABA
11′ ∅BANŠUR AD.KID *da-ga-a-an* *da-a-i* *še-ra-Ø-Ø-Ø*
12′ *Ø-NA* GIŠBANŠUR *še-er* *da-a-i* *še-er* *ar-Ø*
13′ *-zi* *na-at-kán* *an-da* *ú-*
14′ *nu-uš-ši-kán* LÚAZU x
15′ ∅BANŠUR *da-a-i* *na-an-ša-Ø*
16′ *-ma-aš-ša-Ø*

function · verb · realia · agent, deity · place · animal, body, nature · abstract · adjective · foreign · unclear

❷ Count

Each word becomes several features, its form and its sign sequences:

da-a-i

→ word **tai**	0.09
→ signs *da-a-i*	0.09
→ signs *da-a*	0.08
→ signs *a-i*	0.08

A feature weighs more if it is frequent here and rare in the corpus:

$$x_f \propto (1 + \log tf_f) \cdot idf_f$$

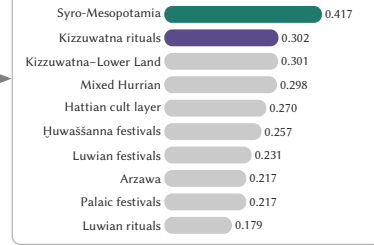
idf of **tai** 1.9; of the sign *an* 1.1

The fragment is a vector $\mathbf{x}$ with 160 of 20,166 features.

❸ Compare

Each tradition is the average vector $\mathbf{c}_t$ of its manuscripts. Similarity is the cosine:

$$s_t = \cos(\mathbf{x}, \mathbf{c}_t) = \sum_f x_f c_{t,f}$$



❹ Decide

The lead of the nearest tradition over the runner-up:

$$m = s_1 - s_2 = 0.42 - 0.30 = 0.12$$

Placed only if $m \geq \tau = 0.028$, the lead at which 80% of test fragments are placed correctly; otherwise left unassigned.

⇒ **Syro-Mesopotamia**

❺ Explain

The lead splits exactly over the features:

$$m = \sum_f x_f (c_{1,f} - c_{2,f})$$

feature	x_f	$c_{1,f}$	$c_{2,f}$	share ×10^3
AZU	0.07	0.13	0.02	+7.8
AZU	0.07	0.12	0.02	+6.8
tai	0.09	0.13	0.05	+6.8
SÍSKUR	0.13	0.08	0.03	+6.6
LÚ-AZU	0.07	0.11	0.01	+6.5
da-a-i	0.09	0.10	0.04	+5.1
ga	0.05	0.02	0.06	−1.6
ke	0.08	0.03	0.05	−1.6

c_1 Syro-Mesopotamia, c_2 Kizzuwatna rituals; bold: word form, regular: sign sequence; italics: Hittite and Akkadograms, upright: Sumerograms.

❻ Check by hand

1′ x-*a-ḫi*-
2′ NINDA.SIG x x
3′ EN SÍSKUR-*ma* *AN*
4′ LÚSIMUG-*ia* *ku-e* x
5′ LÚḪAL *ša-ra-a* *da-a-i*
6′ x$^{\text{HLA}}$-*ma-aš-ša-an* *ku-i-e-eš* *A-NA* GIŠBANŠUR
7′ x *da-aš-ke-ez-zi* *na-at-kán* *ḫa-aš-ši-i* *zi-Ø-Ø-Ø-Ø*
8′ *-in-ni* *zi-ik-ke-ez-zi* EN SÍSKUR-*ma*
9′ *na-an-kán* EN SÍSKUR *A-NA* D*IŠTAR* *ši-pa-Ø-Ø*
10′ *šu-up-pa* *ḫu-u-i-šu* *da-an-zi* UZUGABA
11′ ∅BANŠUR AD.KID *da-ga-a-an* *da-a-i* *še-ra-Ø-Ø-Ø*
12′ *Ø-NA* GIŠBANŠUR *še-er* *da-a-i* *še-er* *ar-Ø*
13′ *-zi* *na-at-kán* *an-da* *ú-*
14′ *nu-uš-ši-kán* LÚAZU x
15′ ∅BANŠUR *da-a-i* *na-an-ša-Ø*
16′ *-ma-aš-ša-Ø*

Darker: larger share of the lead.

AZU, ***tai***, **BANŠUR** decide most. Of the five nearest single manuscripts, 3 are CTH 701, starting with KBo 59.165: leads for the philologist.